

Jackson Zhang

Embedded Systems & IoT Firmware Engineer • 5 Years Experience • MERIQ Index: 71% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: PCB Debugging, FreeRTOS, C/C++, MQTT, Embedded Linux, I2C/SPI/UART, Python

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core PCB Debugging service, reducing latency by 57%.
- Led a squad of 6 engineers delivering scalable microservices processing 13M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 68%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2021 - 2023)

- Designed and maintained FreeRTOS data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using C/C++ saving \$23k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, FreeRTOS, C/C++, MQTT]
Engineered end-to-end production solution leveraging PCB Debugging and FreeRTOS with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [C/C++, MQTT, Embedded Linux, I2C/SPI/UART]
Architected high-resilience system integrating C/C++ with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Robotics and Autonomous Systems

University of Cambridge — 2017 - 2021 (Score: 3.54 / 4.0)